

CS498 Week 07-12

March 2026

Outline

Move from web tooling into core offensive tools such as network discovery (nmap), vulnerability research (CVE/NVD/Exploit-DB + prioritization), exploitation workflow (Metasploit), and privilege escalation fundamentals. These weeks build the technical confidence students need before the “simulated pentest” capstone.

Network Reconnaissance

This section focuses on identifying hosts/services and interpreting scan results professionally. Students learn scan types, service/version detection, common ports, and how to turn “open port + service banner” into a prioritized investigation plan.

Material

- [TryHackMe: Section 5 - Section 7](#)
- [TryHackMe: Bandit network \(Required\)](#)
- [YouTube: Bandit network walkthrough](#)
- [TryHackMe: More nmap \(recommended\)](#)
- [TryHackMe: Nmap Live Host Discovery \(recommended\)](#)
- [nmap: Reference Guide](#)
- [nmap: Options Summary](#)
- [YouTube: nmap tutorial playlist \(For those who like videos\)](#)

Vulnerability Research

This section teaches students how to go from “service & version” to “is this exploitable in the real world?” Students practice reading advisories, understanding CVSS, using vulnerability databases, and documenting what they find (even if they do not exploit).

Material

- [CVE Program: Overview \(Watch Video\)](#)
- [NIST NVD \(National Vulnerability Database\)](#)
- [Exploit-DB \(public exploit archive\)](#)
- [FIRST: CVSS specification \(risk scoring reference\)](#)

Optional Applied Material

If you complete this section early, I advise attempting the following. It will prepare you for the capstone. If you get stuck, reference the tutorial. Comprehend what is happening and where this step fits into the overall penetration testing cycle.

After completing, fill out your LaTeX formatted report. There are example PDFs for each walkthrough tutorial. Try not to use these until the end. I am fine with copying the trivial information (Confidentiality Statement, Contact Info, Assesment Overview), however, fill out the important information yourself to prepare for industry standards. This is the Scope, Executive Summary, Findings Severity Ratings, Security Strengths, Security Weaknesses, and remediation.

Use the LaTeX formatted report from Week 1. Conduct recon, find vulnerabilities, research CVEs, exploit, collect evidence, justify the CVSS score evidence. Don't copy from the tutorial, try to find the vulnerability yourself. Use the appendix section to reference large outputs of evidence. Create clear remediation steps, and an executive summary. Once complete send the document through Discord. If you have any questions please reach out.

- [TryHackMe: Wreath network](#)
- [YouTube: Wreath network walkthrough](#)
- [Example Pentesting Report for Wreath network \(PDF\)](#)

Deliverables

- [Complete Sections 5 & 6 on TryHackMe's "Jr Penetration Tester" Path](#)
- Google Form for Quiz 4
- Complete the [Network Services](#) room & send screenshot to me on Discord.
 - I would recommend practicing making a pentest report for this room
- [Complete Sections 7 on TryHackMe's "Jr Penetration Tester" Path](#)
- Google Form for Quiz 5
- Complete the [Bandit network](#) and submit a [pentesting report \(use as reference\)](#) with three findings. See "Optional Applied Material" for directions. If directions are still ambiguous please reach out.
- If you complete the optional assignment, send it to me through Discord

Quiz 4 Topics

- Scope-aware scanning + safe recon workflow
- Nmap essentials (host discovery, port states, service/version detection, scripts)
- Interpreting results: exposed services, likely attack surface, next steps
- Evidence standards: what outputs/screenshots belong in a report
- Ranking findings with CVSS

Quiz 5 Topics

- Firmware version into CVE lookup (NVD) and validating relevance
- CVSS base metrics (AV/AC/PR/UI/S/C/I/A) then writing a justification
- Prioritizing findings (business impact & exploitability)
- Ranking findings with CVSS